

Confidence Intervals in the AI Use Impact Tracker

A Personal Reference

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1. Why We Use Confidence Intervals

Every metric on the tracker dashboard is a *point estimate* derived from a sample of MHQ survey respondents. The true population value is unknown. A 95% confidence interval (CI) quantifies the sampling uncertainty around that estimate: if we could re-draw the sample many times, ~95% of intervals computed this way would contain the true parameter.

CIs serve three practical purposes in the tracker:

- **Signal vs. noise.** A narrow CI (e.g., ± 0.006) tells us the estimate is stable. A wide CI (± 0.08) warns that the sample may be too small to draw conclusions.
- **Cross-stratum comparison.** If the CIs for two countries do not overlap, the difference is statistically meaningful at the 95% level. If they overlap substantially, we cannot confidently distinguish the two.
- **Trend validation.** A month-over-month shift in WII only qualifies as a real trend if the new estimate falls outside the previous month's CI.

2. Two CI Methods in the Pipeline

The tracker computes CIs using two distinct formulas, chosen based on the statistical nature of the metric.

2.1 Wilson Score Interval (for proportions)

Used for: `adoption_rate`, all `impact_share_*` flags, `positive_impact_share`, `negative_impact_share`.

These metrics are proportions — bounded between 0 and 1 — derived from binary counts (k successes out of n trials). The classical Wald interval ($p \pm z \cdot \sqrt{p(1-p)/n}$) can produce bounds below 0 or above 1, especially when p is near the boundary or n is small. The Wilson score interval corrects this:

$$\begin{aligned} \text{center} &= (\hat{p} + z^2/2n) / (1 + z^2/n) \\ \text{half} &= z \cdot \sqrt{(\hat{p}(1-\hat{p})/n + z^2/4n^2) / (1 + z^2/n)} \\ \text{CI} &= [\max(0, \text{center} - \text{half}), \min(1, \text{center} + \text{half})] \end{aligned}$$

where $\hat{p} = k/n$ is the observed proportion and $z = 1.96$ for 95% confidence.

Key property: The Wilson interval is asymmetric around $\hat{p}$ and always stays within $[0, 1]$, making it the standard choice for proportion CIs in applied statistics.

2.2 Wald (SEM-based) Interval (for continuous means)

Used for: `weighted_impact_index` (the primary dashboard headline metric).

The Weighted Impact Index (WII) is a continuous-valued mean, not a proportion. Each respondent's personal score is the dot product of their binary impact flags and the assigned severity weights, yielding values that range from -1.0 to $+1.5$. Since the metric is unbounded by $[0, 1]$, we use the standard Wald interval based on the standard error of the mean:

$$\text{SEM} = \sigma / \sqrt{N}$$

$$\text{CI} = [\bar{x} - 1.96 \cdot \text{SEM}, \bar{x} + 1.96 \cdot \text{SEM}]$$

where σ is the sample standard deviation (ddof=1), N is the impact denominator count, and $\bar{x}$ is the WII point estimate.

The Wald interval is symmetric around the mean and relies on the Central Limit Theorem. For $N \geq 50$ (our suppression threshold), the CLT approximation is solid.

3. When to Use Which

Metric	Type	CI Method	Rationale
adoption_rate	Proportion	Wilson	Binary: user vs non-user
impact_share_*	Proportion	Wilson	Binary: flag on vs off
positive_impact_share	Proportion	Wilson	Any positive flag set
negative_impact_share	Proportion	Wilson	Any negative flag set
weighted_impact_index	Continuous mean	Wald	Score range -1.0 to $+1.5$

4. Worked Example: Wilson CI for Adoption Rate

Let's walk through the computation for **April 2026, Global** where:

- $N = 16,486$ respondents
- $k = 12,753$ are AI users (frequency ≥ 1)
- $\hat{p} = 12753 / 16486 = \mathbf{0.7736}$

Step-by-step

1. Compute z^2/n :

$$z^2/n = 1.96^2 / 16486 = 3.8416 / 16486 = 0.000233$$

2. Compute the adjusted centre:

$$\text{center} = (0.7736 + 0.000117) / (1 + 0.000233) = 0.77372 / 1.000233 = 0.77354$$

3. Compute the half-width:

$$\begin{aligned} \text{half} &= 1.96 \cdot \sqrt{(0.7736 \times 0.2264 / 16486 + 3.8416 / (4 \times 16486^2)) / 1.000233} \\ &= 1.96 \cdot \sqrt{(0.00001062 + 3.54 \times 10^{-9}) / 1.000233} \\ &= 1.96 \cdot 0.003259 / 1.000233 = 0.006388 \end{aligned}$$

4. Final interval:

$$CI = [0.77354 - 0.00639, 0.77354 + 0.00639] = [0.7672, 0.7799]$$

The pipeline output for this cell: **adoption_rate_ci_low = 0.7653, adoption_rate_ci_high = 0.7781**. (Minor rounding differences from intermediate precision.)

Interpretation: We are 95% confident that the true global AI adoption rate in April 2026 lies between 76.5% and 77.8%. The narrow ± 0.6 pp margin reflects the large sample.

5. Worked Example: Wald CI for Weighted Impact Index

Same cohort: **April 2026, Global**.

The WII is computed over the impact denominator (AI users who provided an impact response).
From the baked data:

- $\bar{x} = WII = +0.1413$
- $CI = [0.1354, 0.1471]$

Reverse-engineering the SEM

From the CI bounds we can recover the SEM:

$$\text{margin} = (0.1471 - 0.1354) / 2 = 0.00586$$

$$SEM = \text{margin} / 1.96 = 0.00586 / 1.96 = 0.00299$$

And we can infer the sample standard deviation. The impact denominator for this month was approximately $N \approx 9,300$ (those who are AI users and gave an impact answer):

$$\sigma = SEM \times \sqrt{N} = 0.00299 \times \sqrt{9300} \approx 0.00299 \times 96.4 \approx 0.288$$

Interpretation: We are 95% confident the true population WII lies between +0.135 and +0.147. The ± 0.006 margin is tight enough that a shift to, say, +0.120 next month would clearly fall outside this band, signalling a real decline.

6. Using CIs to Compare Groups

6.1 Non-overlapping CIs → Significant difference

Suppose in April 2026:

- **Brazil:** $WII = +0.21$, $CI = [0.175, 0.245]$
- **Germany:** $WII = +0.08$, $CI = [0.042, 0.118]$

The intervals do not overlap (Germany's upper bound $0.118 < \text{Brazil's lower bound } 0.175$), so we can state with $>95\%$ confidence that Brazil's AI impact sentiment is more positive than Germany's.

6.2 Overlapping CIs → Inconclusive

Suppose instead:

- **UK:** $WII = +0.15$, $CI = [0.120, 0.180]$

- **Canada:** WII = +0.13, CI = [0.095, 0.165]

The intervals overlap (UK lower 0.120 < Canada upper 0.165). This does not prove the groups are equal — it means our samples are not large enough to distinguish them confidently. A formal two-sample test could still find significance, but the CI overlap is a quick visual heuristic.

Rule of thumb: Non-overlap of 95% CIs implies significance at roughly $p < 0.01$ (stricter than 0.05). So overlapping CIs don't rule out a real difference — they just mean the quick visual check is inconclusive.

7. Where CIs Appear on the Dashboard

Currently, CIs surface in one place on the v2 dashboard:

- **WII hero card sub-line:** Displays 95% CI: [lo, hi] ($\pm$ margin) below the WII value in single-month view only.

CIs are *not* displayed in rolling-window mode (Last 3 / Last 6 months) because the pooled estimate uses sample-size-weighted means across independent monthly slices. You cannot simply average pre-computed CIs; the correct approach would require access to the raw respondent-level data at render time, which the static baked dashboard does not have.

7.1 Potential Extensions

CIs are computed and stored in the Parquet output for every metric and every stratum cell. They could be surfaced in additional places:

- **Error bars on time-series:** Add vertical whiskers to the WII trend line showing $\pm 1.96 \cdot \text{SEM}$ per month.
- **Country scatter confidence ellipses:** On the adoption-vs-WII scatter, draw horizontal/vertical error bars per country.
- **Country table CI column:** Add a CI column to the sortable country table, helping users spot countries with unreliable estimates (wide CIs from small samples).
- **Demographic bar chart whiskers:** Show CI ranges on the gender/age breakdown bars.

Implementation note: All of these require only front-end changes — the CI fields are already in the embedded JSON. The data pipeline does not need modification.

8. Suppression Threshold and CI Width

The pipeline suppresses any stratum cell with fewer than 50 respondents ($N < 50$). This threshold was chosen partly because of CI behaviour: at $N = 50$, the Wilson CI for a proportion of 0.50 is approximately $\pm 14\text{pp}$, and the Wald CI for a mean with $\sigma = 0.3$ is approximately ± 0.083 . Below $N = 50$ the intervals become so wide that the estimates carry little practical value.

Approximate CI half-widths at different sample sizes (for a proportion $p = 0.50$):

N	Wilson $\pm$ ($p=0.50$)	Wald $\pm$ ($\sigma=0.30$)	Usable?
30	± 0.182 (18.2pp)	± 0.107	Suppressed
50	± 0.139 (13.9pp)	± 0.083	Threshold

200	± 0.069 (6.9pp)	± 0.042	Good
1,000	± 0.031 (3.1pp)	± 0.019	Excellent
16,000+	± 0.008 (0.8pp)	± 0.005	Global-level

At our typical global N of ~16K–34K per month, CIs are extremely tight. The real value of CIs shows up at the country or country×gender×age_band level where cells can drop near the suppression boundary.

9. Implementation Reference

The CI computations live in a single file:

- `tracker/src/pipeline/metrics.py`

Wilson CI function (line 38)

`wilson_ci(k, n, z=1.96)` — vectorised NumPy implementation. Takes arrays of success counts and trial counts. Returns (lo, hi) arrays clipped to [0, 1].

Wald CI for WII (lines 131–134)

Computed inline: `mean ± 1.96 * std(ddof=1) / sqrt(N)`. Only generated when $N > 1$.

Where CIs are stored

Every CI appears as two adjacent columns in the Parquet output: `<metric>_ci_low` and `<metric>_ci_high`. These propagate into the baked dashboard JSON and are available in every stratum-level file (global, country, gender, age_band, and all cross-strata).